Homework 4 Theory Questions

Problem 1:

1. Passing by reference is preferred for objects because it avoids the overhead of copying the entire object, which can be large and slow. It also allows the function to modify the original object if needed.
2. Instance members belong to a specific object and hold data unique to it. Static members belong to the class itself, meaning all objects share the same value or function.
3. Multiple-inclusion errors occur when a header file is included more than once in a single translation unit, leading to redefinition errors. You prevent this using include guards (#ifndef, #define, #endif) or #pragma once.
4. Data field encapsulation hides the internal state of an object, protecting it from unauthorized access and modification. This improves maintainability and allows the internal implementation to change without affecting external code.

Problem 2:

1. function overloading
2. dot operator
3. private